

# Econ 101A

## Section 8 (Solutions)

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Everything so far has been abstract: two goods  $x$  and  $y$ . This section puts the same machinery to work on two real decisions. How many hours to work, and how much to consume today versus tomorrow. Nothing new in the math. Each is a constrained utility maximization with a cleverly written budget. The only trick is reading the budget correctly and knowing what plays the role of a price.

### 1 Labor Supply: Consumption versus Leisure

You have a fixed time endowment  $T$  (hours in a day, say). Split it between leisure  $\ell$  and work  $h = T - \ell$ . Consumption  $c$  has price 1, the wage is  $w$ , and you have non-labor income  $M \geq 0$ .

**The budget.** You spend on consumption what you earn plus your non-labor income:

$$c = M + wh = M + w(T - \ell) \quad \Longleftrightarrow \quad \underbrace{c + w\ell}_{\text{spending}} = \underbrace{M + wT}_{\text{full income}}.$$

Read the right-hand form as an ordinary budget. **The wage  $w$  is the price of leisure:** every hour you do not work costs you  $w$  in forgone earnings. The right side,  $M + wT$ , is “full income,” the value of your time if you sold all of it, plus non-labor income.

**Interior optimum.** Maximize  $u(c, \ell)$  subject to  $c + w\ell = M + wT$ . Same tangency as always:

$$\frac{u_\ell}{u_c} = w.$$

The rate at which you trade leisure for consumption equals the wage.

**The no-work corner.** If the interior solution wants  $\ell > T$ , that is infeasible: you cannot enjoy more leisure than you have. Then  $\ell^* = T$ ,  $h^* = 0$ ,  $c^* = M$ . There is a **reservation wage**: below it, work is not worth it and hours are zero.

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**Comparative statics on hours.** Two clean signs and one ambiguous one.

- $\partial h^*/\partial M < 0$ : more non-labor income, if leisure is normal, means you buy more leisure and work less.
- $\partial h^*/\partial w$  is **ambiguous**. A higher wage raises the price of leisure (substitution effect: work more) but also makes you richer (income effect: buy more leisure, work less). When the income effect wins at high wages you get a **backward-bending** labor supply curve.

## 2 Intertemporal Choice: Today versus Tomorrow

Two periods,  $t = 0, 1$ . Incomes  $M_0, M_1$ . Consumption  $c_0, c_1$ , each priced at 1 in its own period. You can save or borrow at rate  $r$ .

**The budget.** Push everything to present value. A dollar tomorrow is worth  $1/(1+r)$  today, so

$$c_0 + \frac{c_1}{1+r} = M_0 + \frac{M_1}{1+r} \equiv W.$$

**The price of future consumption is  $1/(1+r)$ :** to consume one more unit tomorrow you must set aside  $1/(1+r)$  today.  $W$  is lifetime wealth in present value.

**Preferences and the Euler equation.** Utility is  $U(c_0) + \frac{1}{1+\delta}U(c_1)$ , where  $\delta \geq 0$  is the rate of time preference (impatience) and  $U' > 0$ ,  $U'' < 0$ . The Lagrangian gives  $U'(c_0) = \lambda$  and  $\frac{U'(c_1)}{1+\delta} = \frac{\lambda}{1+r}$ . Divide to eliminate  $\lambda$ :

$$\boxed{U'(c_0) = \frac{1+r}{1+\delta} U'(c_1)} \quad (\text{the Euler equation}).$$

It balances the marginal utility of a dollar spent today against a dollar saved and spent tomorrow.

**The benchmark  $r = \delta$ .** Then  $U'(c_0) = U'(c_1)$ , and since  $U'$  is strictly decreasing,  $c_0 = c_1$ : **perfect consumption smoothing**. Saving is  $s = M_0 - c_0$ . You save when income is front-loaded and borrow when it is back-loaded, to flatten the path.

## 3 Exercises

### Exercise 1 (labor supply)

Let  $u(c, \ell) = \ln c + \ln \ell$ , with time endowment  $T$ , wage  $w > 0$ , and non-labor income  $M \geq 0$ .

- Write the budget in full-income form and set up the tangency condition.
- Solve for  $c^*$ ,  $\ell^*$ , and hours  $h^*$ .

- (c) For which wages does the worker supply positive hours? Interpret the threshold.
- (d) Sign  $\partial h^*/\partial M$  and  $\partial h^*/\partial w$ . Is labor supply backward-bending here?

**Solution.** (a) Budget  $c + w\ell = M + wT \equiv F$ . Tangency  $u_\ell/u_c = (1/\ell)/(1/c) = c/\ell = w$ , i.e.  $c = w\ell$ .

(b) This is Cobb–Douglas with equal weights, so each “good” (consumption, and leisure at price  $w$ ) takes half of full income:  $c^* = \frac{F}{2} = \frac{M + wT}{2}$ ,  $w\ell^* = \frac{F}{2}$  so  $\ell^* = \frac{M + wT}{2w}$ . Then

$$h^* = T - \ell^* = T - \frac{M + wT}{2w} = \frac{wT - M}{2w} = \frac{T}{2} - \frac{M}{2w}.$$

(c)  $h^* > 0 \iff wT > M \iff w > M/T$ . The reservation wage is  $M/T$ : if the wage is below the value of your time implied by your non-labor income, you do not work.

(d)  $\frac{\partial h^*}{\partial M} = -\frac{1}{2w} < 0$  (more non-labor income, less work).  $\frac{\partial h^*}{\partial w} = \frac{M}{2w^2} > 0$  (higher wage, more work). With Cobb–Douglas the substitution effect always wins, so labor supply slopes up and is *not* backward-bending. Backward-bending requires preferences where the income effect can dominate at high wages.

## Exercise 2 (intertemporal choice, log utility)

Let  $U(c) = \ln c$ , so lifetime utility is  $\ln c_0 + \frac{1}{1+\delta} \ln c_1$ , with budget  $c_0 + \frac{c_1}{1+r} = W$  where  $W = M_0 + \frac{M_1}{1+r}$ .

- (a) Write the Euler equation and solve for  $c_0^*$  and  $c_1^*$ .
- (b) Evaluate at  $M_0 = 100$ ,  $M_1 = 50$ ,  $r = \delta = 0$ . Find consumption and saving. Is she a saver or borrower, and why?
- (c) Holding  $r$  fixed, sign  $\partial c_0^*/\partial \delta$  and interpret.

**Solution.** (a) Euler:  $\frac{1}{c_0} = \frac{1+r}{1+\delta} \frac{1}{c_1}$ , i.e.  $c_1 = \frac{1+r}{1+\delta} c_0$ . This is Cobb–Douglas in  $(c_0, c_1)$  with the price of  $c_1$  equal to  $1/(1+r)$  and weights 1 and  $\beta = 1/(1+\delta)$ . Spending on  $c_0$  is a share  $\frac{1}{1+\beta}$  of wealth:

$$c_0^* = \frac{W}{1+\beta} = W \frac{1+\delta}{2+\delta}, \quad c_1^* = \frac{\beta}{1+\beta} (1+r)W = \frac{(1+r)W}{2+\delta}.$$

(b) With  $r = \delta = 0$ :  $\beta = 1$ ,  $W = 100 + 50 = 150$ . So  $c_0^* = 150/2 = 75$  and  $c_1^* = 75$ . Saving  $s = M_0 - c_0^* = 100 - 75 = 25$ . She **saves** 25: income is front-loaded ( $M_0 > M_1$ ) and with  $r = \delta$  she wants a flat path  $c_0 = c_1$ , so she moves resources from period 0 to period 1.

(c)  $c_0^* = W \frac{1+\delta}{2+\delta}$ , and  $W$  depends on  $r$ , not  $\delta$ , so

$$\frac{\partial c_0^*}{\partial \delta} = W \frac{(2+\delta) - (1+\delta)}{(2+\delta)^2} = \frac{W}{(2+\delta)^2} > 0.$$

More impatience raises consumption today and lowers saving. Intuitive: a higher  $\delta$  discounts the future more, so you shift consumption toward the present.

### Exercise 3 (intertemporal comparative statics with the IFT)

Keep general  $U$  with  $U' > 0$ ,  $U'' < 0$ , and utility  $U(c_0) + \frac{1}{1+\delta}U(c_1)$ . Substitute the budget as  $c_1 = M_1 + (1+r)(M_0 - c_0)$ , so saving is  $s = M_0 - c_0$ .

- (a) Write the optimality condition as  $F(c_0; r) = 0$  and confirm  $F_{c_0} \neq 0$ .
- (b) Use the implicit function theorem to sign  $\partial c_0^* / \partial r$  for a *borrower* ( $s < 0$ ).
- (c) Explain why the sign is ambiguous for a *saver* ( $s > 0$ ).

**Solution.** (a) The Euler equation with  $c_1$  substituted is

$$F(c_0; r) = U'(c_0) - \frac{1+r}{1+\delta} U'(M_1 + (1+r)(M_0 - c_0)) = 0.$$

Differentiating in  $c_0$  (note  $\partial c_1 / \partial c_0 = -(1+r)$ ):

$$F_{c_0} = U''(c_0) + \frac{(1+r)^2}{1+\delta} U''(c_1) < 0,$$

strictly negative since  $U'' < 0$ , so the IFT applies.

(b)  $\frac{\partial c_0^*}{\partial r} = -\frac{F_r}{F_{c_0}}$ . Here  $\partial c_1 / \partial r = (M_0 - c_0) = s$ , so

$$F_r = -\frac{1}{1+\delta} U'(c_1) - \frac{1+r}{1+\delta} U''(c_1) s = -\frac{1}{1+\delta} \left[ U'(c_1) + (1+r) U''(c_1) s \right].$$

Since  $F_{c_0} < 0$ ,  $\text{sign}(\partial c_0^* / \partial r) = \text{sign}(F_r)$ . For a borrower  $s < 0$ : the term  $(1+r)U''(c_1)s > 0$  (negative times negative), so the bracket is positive and  $F_r < 0$ . Hence  $\partial c_0^* / \partial r < 0$ : a higher rate makes borrowing more expensive and makes the borrower poorer, both pushing today's consumption down. (c) For a saver  $s > 0$ : now  $(1+r)U''(c_1)s < 0$ , so the bracket is  $U'(c_1)$  (positive) plus a negative term, and its sign is not determined by primitives. Economically, substitution pushes  $c_0$  down (saving pays more), but the saver is now richer (income effect pushes  $c_0$  up), and either can dominate.